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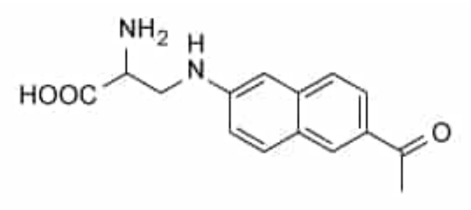


Figure 1 Structure of Anap

The researchers used Anap to biochemically characterize Protein **X**, a novel protein. Protein **X**'s sequence was modified to include both an N-terminal FLAG-tag for purification and a single amber stop codon UAG in the middle of its sequence to code for Anap incorporation. The modified **X** sequence, as well as the sequences encoding an appropriate tRNA and aminoacyl-tRNA synthetase, were cotransformed into *Escherichia coli* for protein expression. Protein **X** was purified, and the λ_{max} of fluorescence emission was measured at varying concentrations of ligand **L**, producing the sigmoidal curve shown in Figure 2.

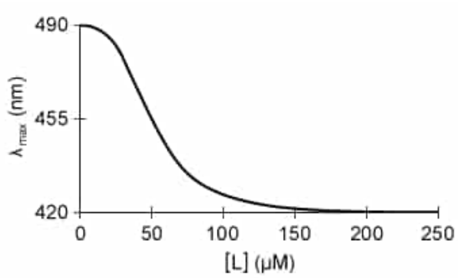


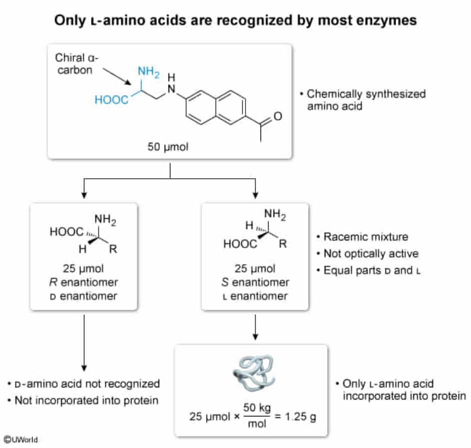
Figure 2 Fluorescence emission maximum of Anap mutant of Protein **X** at various **L** concentrations (Note: K_d of wildtype for **L** = 50 μM)

If the molecular weight of the Protein **X** polypeptide is 50.0 kDa, what is the theoretical maximum amount of full-length Protein **X** that could be produced by adding 50.0 μmol of the Anap mixture obtained from the HPLC column to a culture of *E. coli*? (Note: Assume all other amber stop codons of the *E. coli* genome have been removed via mutation.)

- A. 0.50 g [13%]
- ✔ B. 1.25 g [27%]
- C. 2.50 g [51%]
- D. 5.00 g [7%]

Correct
26% answered correctly

Explanation:



Amino acids are the building blocks of proteins. The basic structure of an amino acid consists of a carbon (known as an α-carbon) bonded to four substituents: a hydrogen, an amino group, a carboxylate group, and a variable R group (also known as the side chain). Apart from glycine, the side chain of which is a second hydrogen atom, **all natural amino acids** are chiral because all four substituents on the α-carbon are distinct. The unnatural amino acid Anap is also chiral because its side chain is distinct from the other three α-carbon substituents.

Although the use of enzymes allows for stereospecific biochemical synthesis of L-amino acids, chemical synthesis (eg, Gabriel, Strecker) produces a mixture of L- and D-amino acids. The passage confirms that the synthesis produced a **racemic mixture** (ie, 50% L-Anap and 50% D-Anap) because the mixture showed no optical activity.

Most enzymes, including aminoacyl-tRNA synthetases and the catalytic domains of ribosomes, recognize only L-amino acids and the proteins that contain them. Consequently, only the 25.0 µmol of L-Anap (*half* of the total 50.0 µmol of Anap) can be used to synthesize Protein **X**. The other 25.0 µmol are D-Anap and cannot be incorporated into the protein. The question states that Protein **X**'s mass is 50.0 kDa, which means its molar mass is 50.0 kg/mol (1 Dalton corresponds to 1 g/mol, so 50 kDa corresponds to 50 kg/mol). The total theoretical yield can then be calculated:

$$25.0\ \mu\ \text{mol L-Anap} \times \frac{1\text{mol Protein } \mathbf{X} \text{ polypeptide}}{1\text{mol L-Anap}} \times \frac{50.0\text{kg}}{\text{mol}} = 1.25\text{g Protein } \mathbf{X} \text{ polypeptide}$$

From a racemic mixture of 50.0 µmol Anap, **25.0 µmol of L-Anap will produce 1.25 g of Protein X.**

(Choice A) 0.50 g would be calculated if 25.0 (the number of micromoles of L-Anap) was *divided* by 50.0 (the molecular weight in kDa) instead of multiplied.

(Choice C) 2.50 g would be calculated if all Anap molecules (both the L- and D-isoforms) were incorporated into Protein **X**. However, only *half* of the total Anap (the L-enantiomer) can be incorporated into protein.

(Choice D) 5.00 g would be calculated if the number of moles of total Anap were *doubled* instead of halved.

Educational objective:
Chemical synthesis of α-amino acids produces a mixture of L- and D-amino acids. However, only the L-isoform can be recognized by most enzymes.

Time spent:0 sec
Subject:
Biochemistry

QID:404104
System:
Amino Acids and Proteins

During protein synthesis, a triplet codon in mRNA is recognized by tRNA anticodons, ultimately resulting in the translation of the nucleic acid sequence into an amino acid sequence. Of the 64 possible three-base permutations, 61 have a highly conserved relationship to particular aminoacyl-tRNAs. The other three—UAG, UGA, and UAA—are stop codons with no corresponding tRNA. Instead, these codons recruit release factors, resulting in protein synthesis termination.

Taking advantage of this, researchers have begun coupling unnatural amino acids, which can contain chemically unique labels, to tRNAs that can recognize the amber stop codon UAG. By mutating the amber stop codon into a protein's mRNA sequence, a site-specific label is incorporated into full-length proteins.

Researchers chemically synthesized Anap, an unnatural amino acid with an acetylnaphthalene derivative as a side chain (Anap, Figure 1). They then confirmed the purity of the synthesized Anap by high-performance liquid chromatography (HPLC). Subsequent measurements using plane-polarized light yielded no optical activity, indicating that the Anap existed in a racemic mixture of both L-Anap and D-Anap. The Anap side chain is fluorescent, and its fluorescence emission spectrum varies depending on its environment. The λ_{max} of free Anap is 490 nm in water but 420 nm in ethyl acetate.

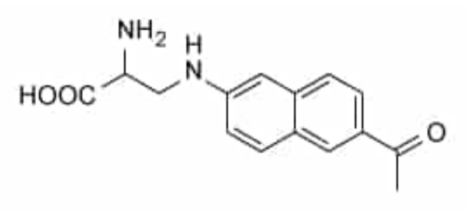


Figure 1 Structure of Anap

The researchers used Anap to biochemically characterize Protein **X**, a novel protein. Protein **X**'s sequence was modified to include both an N-terminal FLAG-tag for purification and a single amber stop codon UAG in the middle of its sequence to code for Anap incorporation. The modified **X** sequence, as well as the sequences encoding an appropriate tRNA and aminoacyl-tRNA synthetase, were cotransformed into *Escherichia coli* for protein expression. Protein **X** was purified, and the λ_{max} of fluorescence emission was measured at varying concentrations of ligand **L**, producing the sigmoidal curve shown in Figure 2.

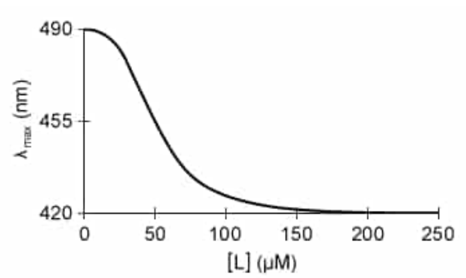


Figure 2 Fluorescence emission maximum of Anap mutant of Protein **X** at various **L** concentrations (Note: K_d of wildtype for **L** = 50 μM)

To purify Protein **X**, the FLAG peptide (shown below) was used to elute the protein from the affinity column.

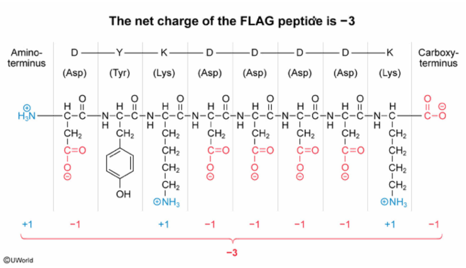
D–Y–K–D–D–D–D–K

What is the net charge of this peptide at pH 7?

- A. −5 [4%]
- ✔ B. −3 [82%]
- C. +2 [8%]
- D. +3 [4%]

Correct
82% answered correctly

Explanation:



The net charge of a peptide can affect many of the peptide's properties, including solubility, ability to be purified by ion-exchange chromatography, and its electrostatic interactions with other molecules. The net charge of a peptide can be estimated by determining the predominant ionization state and charge of each of the peptide's ionizable functional groups and summing those charges.

The FLAG peptide (DYKDDDDK) is an octapeptide (ie, eight amino acids) sequence with 10 ionizable functional groups: the N-terminus, the C-terminus, and each of the eight residue side chains. At a pH of 7, the charges of each individual group can be estimated.

- The N-terminus is positively charged, contributing to the peptide a charge of **+1**.
- Each aspartate (Asp, D) residue is **negatively charged**. The FLAG peptide contains five Asp residues, contributing to the peptide a summed charge of **−5**.
- Tyrosine (Tyr, Y) is ionizable, but is **neutral** at pH 7. Tyrosine contributes to the peptide a charge of **0**.
- Each lysine (Lys, K) residue is positively charged. The FLAG peptide contains two Lys residues, contributing to the peptide a summed charge of **+2**.
- The C-terminus is negatively charged, contributing to the peptide a charge of **−1**.

The overall net charge of the FLAG peptide can be estimated as the sum of each of the individual charges. Therefore, the net charge of FLAG is:

+1 (N-terminus) − 5 (Asp residues) ± 0 (Tyr) + 2 (Lys residues) − 1 (C-terminus)

+3 (Total positive charges) − 6 (Total negative charges)

−3 (Overall net charge)

(Choice A) −5 is the total charge contributed by the Asp sidechains only.

(Choice C) +2 is the total charge contributed by the Lys sidechains only.

(Choice D) +3 is the correct magnitude, but opposite sign, of the correct net charge.

Educational objective:
The overall net charge of a peptide can be estimated by adding the individual charges of the predominant ionization state of each ionizable functional group within the peptide.

Time spent:0 sec
Subject:
Biochemistry

QID:404105
System:
Amino Acids and Proteins

During protein synthesis, a triplet codon in mRNA is recognized by tRNA anticodons, ultimately resulting in the translation of the nucleic acid sequence into an amino acid sequence. Of the 64 possible three-base permutations, 61 have a highly conserved relationship to particular aminoacyl-tRNAs. The other three—UAG, UGA, and UAA—are stop codons with no corresponding tRNA. Instead, these codons recruit release factors, resulting in protein synthesis termination.

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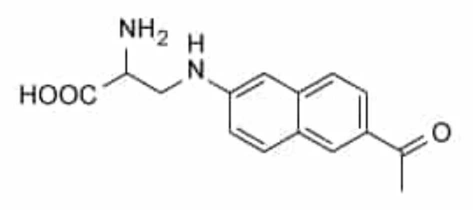

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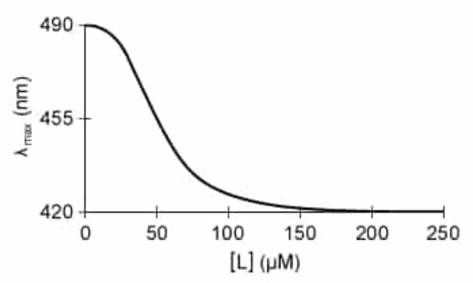


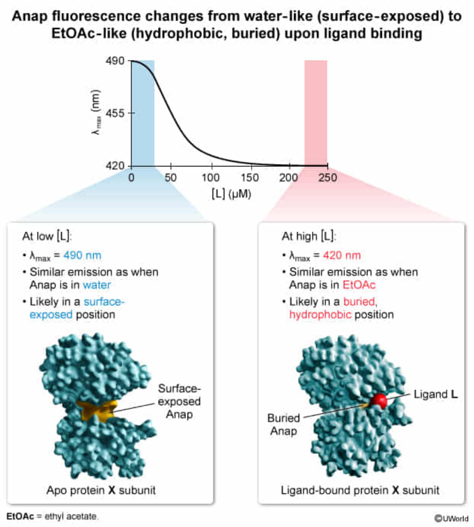
Figure 2 Fluorescence emission maximum of Anap mutant of Protein **X** at various **L** concentrations (Note: K_d of wildtype for **L** = 50 μM)

Which of the following conclusions is most strongly supported by the data in the passage?

- A. The degeneracy of the three stop codons can be explained by the wobble base pair. [15%]
- B. The site chosen for mutation to Anap neither affects nor is affected by ligand binding. [18%]
- ✔ C. The Anap residue moves from a surface-exposed to a buried position upon ligand binding. [60%]
- D. The genetic code is absolutely conserved among natural life on Earth. [5%]

Correct
61% answered correctly

Explanation:



Protein folding is driven largely by the hydrophobic effect. As a result, the buried **core residues** of soluble proteins tend to be **hydrophobic** while the **surface-exposed residues tend to be hydrophilic**. **Conformational changes** (eg, as a result of ligand binding or of posttranslational modifications) alter the three-dimensional structure of a protein. A conformational change can cause individual residues to move from surface-exposed to buried positions, or vice versa.

The passage states that Anap is a fluorescent amino acid with a maximum emission wavelength λ_{max} that changes in different environments. Figure 2 shows that the Protein **X** Anap mutant displays λ_{max} of **490 nm with no ligand bound**, which matches the λ_{max} of Anap in water, indicating that Anap is on the surface of the unbound (ie, apo) Protein **X** where its sidechain is exposed to water.

Figure 2 also shows that the Protein **X** Anap mutant displays a λ_{max} of **420 nm when saturated with ligand**. The conformational change upon ligand binding alters the environment of Anap such that its λ_{max} now matches that of free Anap in ethyl acetate, which is much less polar than water. Therefore, Anap is likely positioned toward the more hydrophobic, buried core in ligand-bound Protein **X**.

Together, these data support the conclusion that **the Anap residue moves from a surface-exposed to a buried position upon ligand binding**.

(Choice A) The wobble base pair partially explains why multiple codons encode the same amino acid at the *third base of a codon*, but the UGA stop codon differs from the other two stop codons at its *second* base.

(Choice B) Although the introduction of an Anap residue does not affect the K_d of Protein **X** for ligand **L**, ligand **L** *does affect* the Anap residue as shown by its changing λ_{max}.

(Choice D) Although the genetic code is highly conserved, some organisms do have slight changes. The passage shows that researchers *can* alter the code (by changing a stop codon into a coding codon), demonstrating that genetic code alteration, although rare, is possible.

Educational objective:
Most soluble proteins have a hydrophilic surface and a hydrophobic core. Conformational changes that rearrange a residue from surface-exposed to buried also change the residue's environment from hydrophilic to hydrophobic.

Time spent:0 sec
Subject:
Biochemistry

QID:404106
System:
Amino Acids and Proteins

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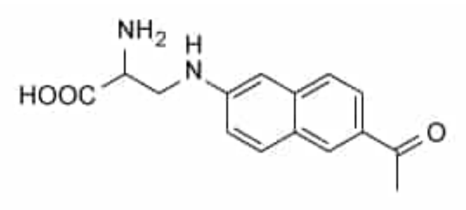

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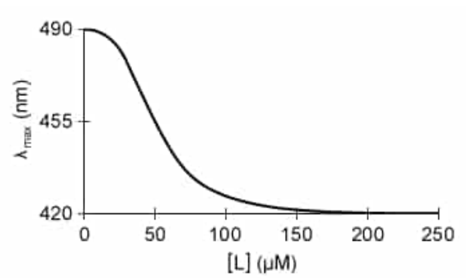


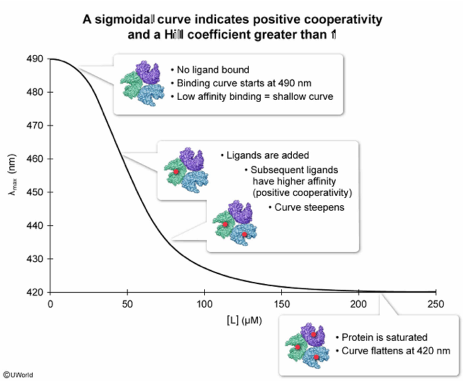
Figure 2 Fluorescence emission maximum of Anap mutant of Protein **X** at various **L** concentrations (Note: K_d of wildtype for **L** = 50 μM)

Based on the data shown in Figure 2, Protein **X** can best be described as:

- A. a positively cooperative protein with a Hill coefficient less than 1. [20%]
- B. a negatively cooperative protein with a Hill coefficient exactly equal to 1. [19%]
- ✔ C. a positively cooperative protein with a Hill coefficient greater than 1. [46%]
- D. a noncooperative protein. [12%]

Correct
47% answered correctly

Explanation:



In proteins, cooperativity occurs if the binding of one ligand influences the binding of subsequent ligands. The Hill coefficient is a quantitative measure of cooperativity. When binding at one site makes binding of subsequent ligands *easier*, the interaction is an example of **positive cooperativity**, and the **Hill coefficient has a value greater than 1**. Graphically, positive cooperativity of ligand binding can be recognized by the appearance of an S-shaped sigmoidal curve as ligand concentration linearly increases.

Figure 2 is essentially a binding curve for Protein **X** that uses the emission wavelength as a reporter for ligand binding. Because the wavelength decreases as the fraction of Protein **X** increases, the binding curve appears inverted; however, the overall shape of the binding curve is still recognizable. At low concentrations of **L**, the curve is shallow, indicating low-affinity binding. The curve steepens as **ligand-induced conformational changes** cause **high-affinity binding** for subsequent ligands. The curve flattens again as the population of Protein **X** becomes saturated with ligand, forming the top of the **sigmoidal** curve.

Because of the sigmoidal binding curve shown in Figure 2, Protein **X** can be described as a protein showing **positive cooperativity** and a **Hill coefficient greater than 1**.

(Choice A) A Hill coefficient less than 1 is indicative of *negative* cooperativity. Negative cooperativity is not represented by a sigmoidal curve.

(Choices B and D) A Hill coefficient equal to 1 is indicative of noncooperativity. Noncooperative binding is represented by a hyperbolic curve.

Educational objective:
Positive cooperativity is indicated on a binding curve by a sigmoidal shape and a Hill coefficient greater than 1.

Time spent: 0 sec	**QID:** 404107
Subject: Biochemistry	**System:** Amino Acids and Proteins